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REVIEW

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Associations of parenting styles with substance use in the offspring—A systematic review and meta-analysis

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Abstract

Issues: Parenting styles have been suggested to predict the use of psychoactive substances in the offspring, although the size of associations might vary between cultures. The present meta-analysis tested whether parenting styles show concurrent and longitudinal associations with substance use and whether this association is moderated by study characteristics.

Approach: A systematic search in electronic data bases resulted in 184 studies that were included in multi-level meta-analysis.

Key Findings: An authoritative parenting style correlated with lower substance use ($r = -0.12$, confidence interval [CI] -0.14 to -0.10) while the reverse was found for neglectful ($r = 0.10$, CI 0.08 to 0.13), permissive ($r = 0.04$, CI 0.01 to 0.07) and authoritarian parenting ($r = 0.03$, CI 0.00 to 0.05). Analysis of cross-lagged effects found only an effect of authoritative parenting on decline of substance use ($r = -0.06$, CI -0.12 to -0.00). The size of associations varied, in part, by levels of horizontal and vertical collectivism or individualism, assessment of three versus four parenting styles, type of substance, and use of same versus different informants for assessing parenting and substance use.

Implications: Parenting styles show small to very small associations with substance use.

Conclusions: Although parents are recommended to behave in an authoritative manner, we should have only modest expectations about the effects of parenting styles on substance use in the offspring.

KEYWORDS

adolescents, alcohol or other drugs, authoritative parenting, emerging adults, parenting styles

1 | INTRODUCTION

Substance use in adolescence and beyond is a significant public health concern. For example, adolescent use of alcohol, nicotine and cannabis does negatively affect cognitive and brain development as well as mental health [1]. Early initiation of substance use is a risk factor for later substance use disorders [2]. High substance use also increases the risk for a broad range of physical health conditions and premature mortality [3].

The family, and parenting in particular, has been suggested to play a fundamental role in the prevention and treatment of substance use in young people [4], although many other social factors (e.g., caring peer relations and support from school) and individual factors, such as problem-solving skills, also play a role [5]. While a large

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number of different parenting practices can be distinguished (e.g., autonomy support, monitoring, rule setting, warmth), parenting styles are defined by the combination of core parenting practices. Extending the work of Baumrind [6], Maccoby and Martin [7] distinguished four parenting styles according to the combination of responsiveness/warmth and control/demandingness. The authoritative parenting style is characterised by combining high levels of responsiveness and reasonable control. These parents set clear rules for their children, provide reasons for them and are willing to consider their children's opinions in order to reach an agreement. The authoritarian parenting style is defined by high control combined with low warmth. These parents are strict, less open to compromise and require their children's obedience. A permissive/indulgent parenting style is characterised by high warmth combined with low control. These parents do not want to restrict their children, respect all of their children's decisions and tolerate their behaviours to a maximum degree. Finally, a neglectful (or uninvolved) parenting style is characterised by low levels of warmth and low control. These parents show low interest in their children's life and do not require responsible behaviour from them. An authoritative parenting style is considered as best for the development of the offspring because parental warmth makes the child more receptive to parental influences while parental control and open exchange with the children promote the development of psychosocial competence, such as self-regulation [8]. In contrast, neglectful parenting is considered as most developmentally inappropriate [7].

With regard to associations of parenting styles with substance use in the offspring, three narrative reviews reported some evidence for authoritative parenting being associated with below-average substance use while the reverse was found for neglectful parenting [4, 9, 10]. Inconsistent results have been reported on associations of authoritarian and permissive parenting with substance use. For example, while some studies found elevated levels of substance use in the offspring of permissive parents (e.g., [11, 12]), the reverse has been observed in others [13]. Garcia and Garcia [13] suggested that this heterogeneity reflects variations of cultural appropriateness of parenting, and referred to the different forms of collectivism and individualism proposed by Triandis and Gelfand [14]. These authors distinguished four groups of cultures based on the combination of two dimensions, that is, perception of the self as part of a collective (collectivism) versus autonomous (individualism) and emphasis of equality (horizontal relations) versus acceptance of inequality (vertical relations; see Data S1, Supporting Information). If cultures emphasise equalitarian relations (i.e., horizontal collectivism, such as Spain or Brazil), affective ties (such as parental warmth) rather than behaviours that emphasise hierarchical relations (such as control and strictness) would promote child development, thus indicating that a permissive parenting style (i.e., high warmth plus low control) would be most effective. In contrast, an authoritarian parenting style would be culturally appropriate and effective in vertical collectivistic societies that accept inequality and perceive authoritarian and controlling practices in general as an organisational strategy that fosters harmony within the family. An authoritative parenting style would be most effective in 'authoritative contexts' such as middle-class European-American families that appreciate an autonomous individual and accept some level of inequality such as firm control as socialisation practices (i.e., vertical individualism). The authors did not specify a parenting style that would be most adequate in horizontal individualistic cultures. However, whether associations of parenting styles with substance use vary by national levels of individualism and collectivism has not yet been systematically tested.

Another source of variability of the effect of permissive parenting may be the use of the original Baumrind model of three parenting styles [6] versus the four-styles-model by Maccoby and Martin [7]. The former model defined permissive parenting exclusively by low parental control while the latter model defined permissive parenting by the combination of low control and high warmth. Associations of permissive parenting with child outcomes may be less negative or even positive if permissiveness is defined by the combination of low control and high warmth rather than only by low control.

Available narrative reviews [4, 9, 10] could not provide information on the mean size of associations of parenting styles with substance use. However, this information would be important for evaluating the practical importance of this association. In addition, the reviews addressed only bivariate associations between parenting styles and substance use. Bivariate correlations may reflect effects of parenting styles on substance use, effects of substance use on parenting styles, as well as effects of third variables. Testing cross-lagged effects of parenting styles on change in substance use (and of substance use on change in parenting styles) is, therefore, relevant for the analysis of the direction of effects. Furthermore, available narrative reviews did not analyse whether the size of association of parenting styles with substance use is moderated by study and sample characteristics, such as type of substance, child age and gender.

Arguments for associations of parenting styles with use of psychoactive substances and other outcomes [8] do, in principle, apply to the consumption of all psychoactive substances and other kinds of risky behaviour.

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Nonetheless, the size of associations might differ. For example, associations may be weaker in the case of substances that are consumed only by a small minority (i.e., illegal drugs as compared to alcohol) because most young people will not use these substances regardless of their parents' behaviour.

Parental influences on substance use in their offspring may decline across adolescence and emerging adulthood, for example, due to spending less time together or because substance use is increasingly habit-driven rather than affected by immediate social influences. While support for this assumption has been found in studies on associations of parenting practices, such as monitoring, with substance use [15], age differences in associations of parenting styles with substance use have not yet been systematically analysed. Gender development theory suggests that relationships with parents may be more salient for girls than boys [16], and a meta-analysis by Lac and Crano [17] found stronger associations of monitoring with young peoples' marijuana use in female-only samples than in other samples. Similarly, as mothers are often the primary caregiver, they might be more influential than fathers. Although Jiménez-Iglesias et al. [18] found stronger associations of substance use with maternal as compared to paternal knowledge about the child's whereabouts, associations of other indicators of monitoring did not vary systematically by parental gender.

Aspects of study quality may also affect the size of associations of parenting styles with substance use. Associations may be stronger if information on parenting and substance use comes from the same source (due to shared methods variance) and if the studies are published rather than unpublished (as nonsignificant results may be less likely published than significant results; 'file-drawer problem', [19]). We were also interested in whether effects vary depending on the response rate (as a low response rate may lead to biased results) and on the use of validated versus non-validated measures of parenting and substance use, although no directed hypothesis could be stated based on the literature.

2 | RESEARCH QUESTIONS

In sum, the present study is the first meta-analysis on associations of parenting styles with substance use in the offspring. In the first research question, we ask for the size of bivariate associations of substance use with the four parenting styles defined by Maccoby and Martin [7]. The second research question addresses whether there are significant cross-lagged effects of parenting styles on change in substance use, and of substance use on change in parenting styles. The third research question addresses whether the size of associations of parenting styles with substance use varies by horizontal/vertical collectivism/individualism, the use of a three- versus four-parenting-styles-model, by type of psychoactive substance, child age and gender, parental gender and indicators of study quality.

3 | METHODS

3.1 | Sample

Studies were identified through electronic databases (PsycInfo, MEDLINE, Web of Science and PSYNDEX [an electronic database of psychological literature from German-speaking countries]), using the combination of search terms (substance use OR alcohol OR tobacco OR nicotine OR cigarettes OR drug OR marijuana) AND (parenting style*). References sections of the identified studies were checked for additional papers.

Studies were included if they meet the following conditions:

a. The studies assessed associations of one or more parenting styles defined by Maccoby and Martin [7] with use of psychoactive substances (alcohol, nicotine, marijuana, other drugs) in the offspring.
b. The substances were not prescribed by a physician.
c. The sample consisted of adolescents and emerging adults (mean age <30 years).
d. Bivariate and/or cross-lagged associations of parenting styles with substance use were reported or could be computed based on the available information.
e. The study was published or made available before March 2024.

Studies were excluded if they:

a. Reported only multivariate effect sizes that statistically controlled for third variables (because these effect sizes cannot be combined with bivariate effect sizes);
b. Had samples older than 30 years (because studies with older samples addressed only retrospective reports on parenting and because risk for biased memories increases with length of time interval [20]);
c. provided only a sum-measure of child outcomes that combined substance use with other behaviours;
d. duplicated the results of included studies; and
e. were not available with interlibrary loan.

In order to include studies from different regions of the world that differ in collectivism/individualism, we did not limit the included studies to those written in English.

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Studies that have not been published in scientific journals or books but were identified in the electronic data bases (e.g., dissertations) were also included in order to reduce the risk for file-drawer problems.

The literature search was completed on 1 March 2024. We identified 673 studies. If the full text was not available, authors were asked for a copy in the case of available contact information. After screening and assessing for eligibility, 184 studies were included in the meta-analysis. A preferred reporting items for systematic reviews and meta-analyses (PRISMA) flow chart for the search of studies is provided in Figure 1. The PRISMA checklist, the list of included studies and selected study characteristics can be found in Data S1. The study has been preregistered at PROSPERO (ID 512911).

The following information was coded from the individual studies: number of adolescents/emerging adults, response rate, mean age, percentage of girls, percentage of members of ethnic minorities, publication status (1 = published, 2 = unpublished), source of parenting (1 = mother, 2 = father, 3 = parental dyad), rater of parenting (1 = child, 2 = parent, 3 = observer, 4 = multi-informant), rater of substance use (1 = child, 2 = parent, 3 = others), type of substances consumed (1 = alcohol, 2 = nicotine/cigarettes, 3 = marijuana, 4 = other/mixed substances), use of validated measure of parenting styles (2 = yes, 1 = no), identical/different data source (1 = information on parenting and substance use from same source, 2 = use of different sources), and the size of bivariate and cross-lagged correlation between parenting styles and substance use in the offspring. Data on national levels of horizontal/vertical collectivism and individualism of 28 countries were taken from reference [21]. We used the Mixed Methods Appraisal Tool [22] for evaluating study quality. A review protocol is available from the first author on request.

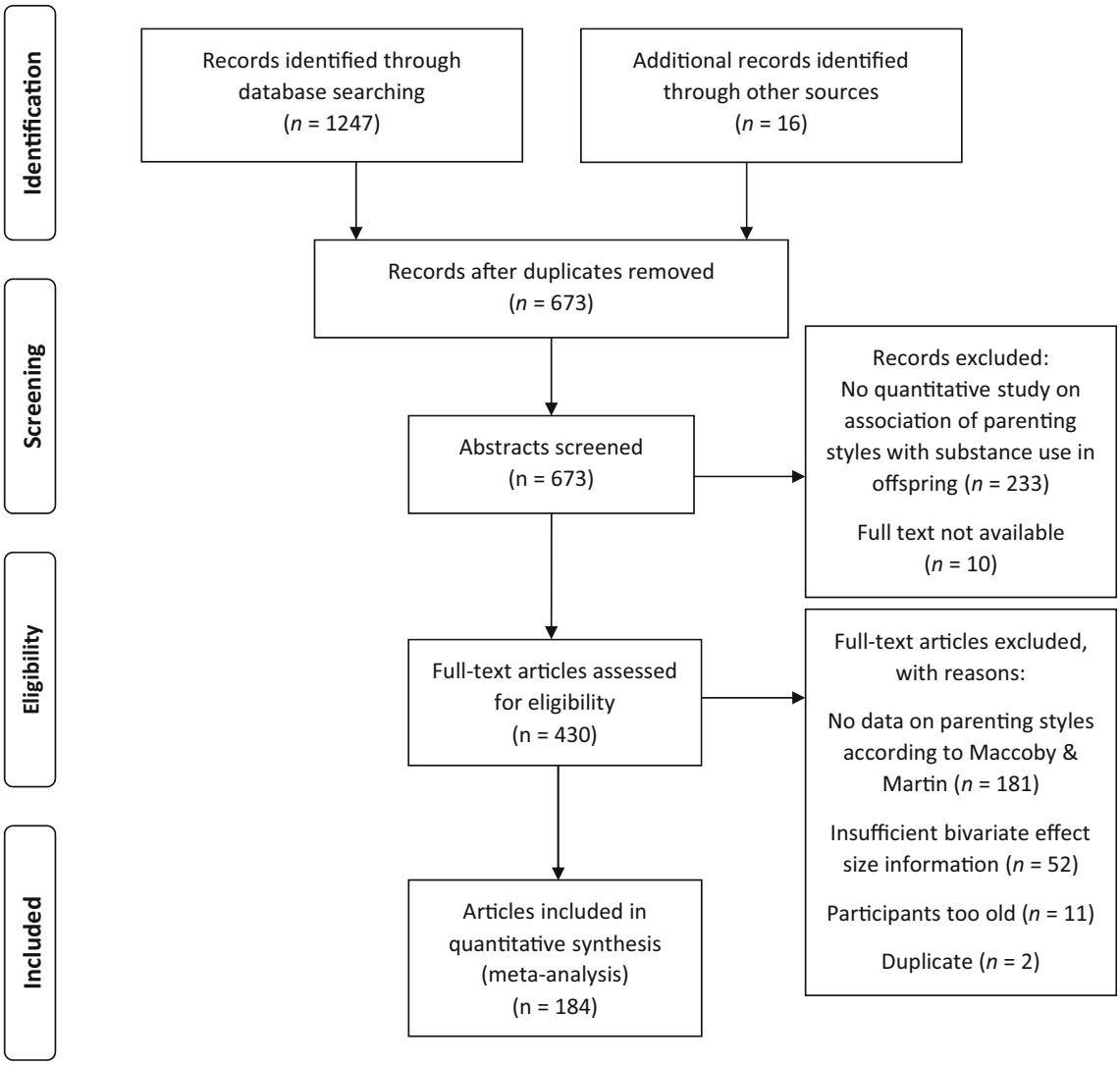

FIGURE 1 PRISMA 2009 flow diagram.

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If associations were provided for several substances consumed by the same sample, these effect sizes were entered separately. Similarly, if data on parenting styles were available from more than one source, effect sizes were coded separately. Search for studies and coding of the studies was conducted independently by the two authors. A mean inter-rater reliability of 94% (range 90%–99%) was established for inclusion and coding of categorical variables and a mean intra-class coefficient of 0.92 for continuous variables. Differences were resolved by discussion.

3.2 | Statistical analysis

First, correlations were transformed using Fisher's *r*-to-*z* transformation. Outliers that were more than two SD from the mean effect size were recoded to the value at two SD, based on recommendations from Lipsey and Wilson [23]. After computing weighted mean effect sizes, the *z*-scores were back-transformed to the original metric of *r*.

As part of the studies provided more than one effect size (mainly due to assessment of use of different substances), the dependency of effect sizes was addressed with multi-level meta-analysis [24]. This approach considers the hierarchic data structure as individual correlations (at level-1) are nested within studies (level-2), which are nested within countries (level-3). Random-effects models were computed that consider random population effects as well as sampling error. For analysing moderator effects of sample or study characteristics, predictors were introduced at level-2 or 3, respectively. Risk for publication bias was addressed by comparing results from published and unpublished studies as well as computing trim-and-fill analysis [25]. Trim-and-fill analysis tests for an asymmetrical distribution of the effect sizes and imputes possibly missing effect sizes. Data were analysed with *metafor* in R [26].

4 | RESULTS

The 184 studies provided data on 273,020 individuals. They had a mean age of 15.08 years (SD = 2.43, range 10.4–27.9); 52.94% were female and 45.08% were members of an ethnic minority. The mean response rate was 74.61%. Twenty-four studies provided longitudinal data; 45 studies were unpublished. Studies from 45 countries were included in the meta-analysis with most studies having been conducted in the United States (70), Brazil (11) and Spain (10). Ninety-seven studies analysed associations of parenting styles with alcohol use, 46 addressed cigarettes and 16 marijuana use. In addition, 69 studies had assessed consumption of other drugs or used a sum-score that combined use of different psychoactive substances. Parenting styles were most often assessed with the Parental Authority Questionnaire ([27]; 34 studies), versions of the Parenting Style Inventory ([28]; 26 studies), and the Authoritative Parenting Measure ([29]; 11 studies). Substance use was most often assessed with questions on the frequency and/or quantity of use (147 studies); 26 studies used screenings for risky use or substance use disorders such as the Alcohol Use Disorders Identification Test ([30]; 22 studies), and 6 studies compared individuals with and without substance use disorders.

Regarding study quality, all studies used convenience samples, but 88% of them had not preselected participants according to criteria that may affect associations of parenting styles with substance use, such treatment-seeking youth. Only about 17% of the studies confirmed a response rate ≥80%; 58% of the studies used validated measures of parenting styles, and no study provided validating information on self-reports of substance use, such as drug screenings or data from a second informant. Complete information for computing associations of parenting styles with substance use was provided in 96% of the included studies.

A three-level meta-analysis was specified which estimated within-study variance (mainly reflecting different substances or parenting by mothers versus fathers), between-study variance within countries, and between-country variance. As shown in Table 1, an authoritative parenting style was associated with lower substance use ($r = -0.12$) while an authoritarian ($r = 0.03$), permissive ($r = 0.04$) and neglectful parenting style ($r = 0.10$) were associated with higher substance use. The statistical significance of the variance components was tested with likelihood-ratio-tests. There was significant within- and between-study variance of the association of the four parenting styles with substance use. However, only the size of the correlation of permissive parenting with substance use varied significantly between countries.

The second research question asked whether parenting styles would predict change in substance use and vice versa. Cross-lagged associations of parenting styles with change in substance use have been less often addressed than concurrent bivariate correlations (Table 1). There was only a statistically significant cross-lagged effect of authoritative parenting on change in substance use ($r = -0.06$), indicating that higher authoritative parenting predicted a very small decline in substance use over time. As only up to two cross-lagged effect sizes of initial substance use on change in parenting styles were available, no meta-analysis was computed.

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TABLE 1 Weighted mean associations of parenting styles with substance use.

Parenting style	k	r	r_{LL}	r_{UL}	t	Q	Level-1 variance	Level-2 variance	Level-3 variance
Bivariate associations									
Authoritative	321	−0.12	−0.14	−0.10	−12.88***	3587.18***	0.0013***	0.0104***	0.0000
Authoritarian	260	0.03	0.00	0.05	2.27*	2533.88***	0.0006***	0.0121***	0.0005
Permissive	266	0.04	0.01	0.07	2.29*	3066.20***	0.0009***	0.0151***	0.0033*
Neglectful	171	0.10	0.08	0.13	8.78***	1213.32***	0.0003***	0.0095***	0.0000
Cross-lagged associations of parenting styles with change in substance use									
Authoritative	22	−0.06	−0.12	−0.00	−2.10*	121.24***	0.0002	0.0063	0.0000
Authoritarian	12	0.02	−0.02	0.05	1.11	13.18	0.0000	0.0000	0.0000
Permissive	12	0.00	−0.04	0.05	0.18	22.22*	0.0013	0.0000	0.0000
Neglectful	17	0.07	−0.02	0.16	1.66	102.26***	0.0021	0.0048	0.0027

Abbreviation: k, number of effect sizes; r, weighted mean correlation; r_{LL}/r_{UL} = lower/upper level of the 95%-confidence interval; t, test of significance of r; Q, test of heterogeneity of effect sizes.
*$p < 0.05$.
***$p < 0.001$.

Given the small number of longitudinal effect sizes, moderator analyses were only computed for bivariate associations of parenting styles with substance use. With regard to the third research question, there were few moderating effects of horizontal and vertical individualism as well as horizontal collectivism (Table 2). Vertical collectivism and vertical individualism moderated the size of the association of authoritative parenting with substance use. Stronger negative correlations of authoritative parenting with substance use were found in countries with higher vertical collectivism and higher vertical individualism scores. For example, a mean correlation of $r = -0.27$ ($p < 0.001$) was found in countries with highest vertical individualism (>6) as compared to $r = -0.04$ ($p < 0.05$) in countries with lowest vertical individualism (<4.2). Similarly, associations were $r = -0.15$ ($p < 0.001$) in countries with highest vertical collectivism (>6) as compared to $r = -0.10$ ($p < 0.01$) in countries with lowest vertical collectivism (<5). In addition, associations of permissive parenting with substance use were less positive in countries with higher horizontal collectivism scores. For example, while a mean weighted correlation of $r = 0.08$ ($p < 0.01$) was found in the countries with lowest horizontal collectivism scores (<6.75), the weighted correlation was $r = -0.01$ ($p < 0.28$) in the countries with the highest scores (>7.8).

Associations of permissive parenting with substance use varied by the conceptualisation of this parenting style ($b = -0.07$, $p < 0.004$). Permissive parenting was associated with higher substance use if studies defined this style exclusively by low parental control ($r = 0.11$, $p < 0.003$) but not in studies that defined this style by the combination of low control and high warmth ($r = 0.00$, $p < 0.88$). We also checked whether studies from countries with high level of horizontal collectivism that measured permissive parenting via low control and high warmth would find hints for a protective effect of permissive parenting. Although we observed a very small negative correlation of permissive parenting with substance use, the effect size was not statistically significant ($r = -0.05$, $p < 0.06$).

There was one moderating effect of the kind of psychoactive substance: associations of authoritarian parenting with tobacco use were stronger (more positive) than associations with alcohol use. In addition, there were stronger associations of substance use with the authoritative and permissive parenting style if data on parenting and substance use came from the same informant. In contrast, there were no moderating effects of child age, child gender, parental gender, response rate, use of a validated parenting measure and publication status. Trim-and-fill analysis indicated that no effect sizes of possibly missing studies had to be imputed, thus indicating no risk for publication bias. Given the lack of variance of the quality of the substance use measures, no moderating effect of this criterion of study quality could be tested.

5 | DISCUSSION

The present meta-analysis found that an authoritative parenting style was, on average, associated with lower substance use in the offspring while the reverse was found with regard to neglectful, permissive and

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TABLE 2 Test for moderating effects of sample and study characteristics.

	Association with authoritative parenting					Association with authoritarian parenting				
Moderator	***b***	**LL**	**UL**	**df**	***t***	***b***	**LL**	**UL**	**df**	***t***
Horizontal collectivism	−0.005	−0.073	0.063	200	−0.15	−0.011	−0.077	0.055	153	−0.32
Vertical collectivism	−0.065	−0.130	−0.001	200	−1.99*	0.043	−0.033	0.125	153	1.12
Horizontal individualism	−0.073	−0.149	0.004	200	−1.88	−0.050	−0.128	0.029	153	−1.25
Vertical individualism	−0.071	−0.117	−0.024	200	−3.00**	−0.010	−0.077	0.057	153	−0.30
Substance: tobacco[a]	0.013	−0.012	0.037	318	1.01	0.028	0.004	0.051	257	2.31*
Substance: marijuana[a]	0.001	−0.031	0.033	318	0.05	0.014	−0.014	0.042	257	0.99
Substance: other/mixed drugs[a]	0.001	−0.024	0.026	318	0.09	0.021	−0.001	0.043	257	1.86
Age	0.002	−0.004	0.007	229	0.55	0.002	−0.005	0.008	197	0.45
Child gender (% female)	0.000	−0.000	0.001	304	0.69	−0.000	−0.001	0.000	245	−1.32
Mother[b]	−0.018	−0.038	0.002	113	−1.77	−0.019	−0.038	0.000	94	−1.96
Validated parenting measure	0.032	−0.003	0.067	320	1.80	−0.041	−0.084	0.003	259	−1.86
Response rate	−0.000	−0.001	0.001	165	−0.28	0.002	−0.005	0.008	200	0.54
Same informant	−0.083	−0.152	−0.014	320	−2.36*	−0.005	−0.079	0.069	259	−0.13
Published	−0.027	−0.068	0.014	320	−1.29	−0.001	−0.048	0.047	259	−0.03

	Association with permissive parenting					Association with neglectful parenting				
Moderator	***b***	**LL**	**UL**	**df**	***t***	***b***	**LL**	**UL**	**df**	***t***
Horizontal collectivism	−0.112	−0.212	−0.011	152	−2.19*	−0.004	−0.080	0.073	90	−0.09
Vertical collectivism	0.003	−0.167	0.172	152	0.03	−0.002	−0.094	0.090	90	−0.05
Horizontal individualism	−0.103	−0.210	0.004	152	−1.90	−0.047	−0.200	0.107	90	−0.60
Vertical individualism	0.065	−0.079	0.210	152	0.89	0.014	−0.063	0.091	90	0.36
Substance: tobacco[a]	−0.012	−0.037	0.013	253	−0.94	−0.012	−0.035	0.012	168	−0.97
Substance: marijuana[a]	−0.013	−0.042	0.017	263	−0.84	−0.007	−0.037	0.023	168	−0.45
Substance: other/mixed drugs[a]	0.007	−0.017	0.031	263	0.56	−0.016	−0.038	0.006	168	−1.47
Age	−0.002	−0.010	0.005	208	−0.53	0.003	−0.005	0.010	140	0.77
Child gender (% female)	0.000	−0.000	0.001	251	0.96	−0.001	−0.001	0.000	161	−1.65
Mother[b]	0.015	−0.008	0.038	94	1.29	0.014	−0.013	0.041	54	1.03
Response rate	−0.001	−0.002	0.001	126	−0.78	−0.001	−0.002	0.001	89	−0.67
Validated parenting measure	−0.024	−0.073	0.026	265	−0.94	−0.038	−0.082	0.008	170	−1.70
Same informant	0.118	0.041	0.196	265	3.01**	−0.044	−0.134	0.046	170	−0.97
Published	−0.004	−0.058	0.059	265	−0.15	0.023	−0.029	0.075	170	0.87

Note: *b*, unstandardised regression coefficient; LL/UL, lower/upper level of the 95% confidence interval of the regression coefficient; *t*, test for significance of the moderator.
*$p < 0.05$.
**$p < 0.01$.
[a]Comparator is alcohol.
[b]Comparator is father.

authoritarian parenting. In longitudinal studies, only an authoritative parenting style predicted change in substance use over time. Results varied, in part, by country-level of horizontal and vertical collectivism, vertical individualism, assessment of parental permissiveness, as well as type of substance and use of the same versus different informants.

The present meta-analysis supports the conclusions of narrative reviews that authoritative parenting is associated with lower substance use in the offspring while the neglectful parenting style relates to higher substance use [4, 9, 10]. It goes beyond these narrative reviews in showing that the associations are, on average, small according to common rules for interpreting effect sizes [31]. It also

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adds to our knowledge on parenting in showing that the authoritarian and permissive parenting style are, on average, related to elevated substance use, although the average size of the association is only very small in statistical terms [31] and the way of assessing permissiveness played a role.

As associations of parenting styles with substance use in the offspring were small or even very small, readers might question the practical significance of parenting styles for preventing or reducing substance use in adolescents and emerging adults. However, as substance use is influenced by different social contexts (e.g., peer group, parents, prevention programs at schools) and personality variables (e.g., outcome expectancies, habits), each of these factors will explain only a limited amount of variance [5]. In addition to general parenting styles, other parental variables also play a role, such as parents' rule setting concerning substance use of their children [32].

Given the negative health effects of substance use [1], small effects on substance use are meaningful. Because we mainly assessed concurrent associations (e.g., with drinking in the last month), we missed effects of parenting at earlier times and in the future, which could lead to an underestimation of parenting effects. In addition, there are subgroups that react more strongly towards positive and negative parenting depending on the collectivistic/individualistic values in the society [13] or on the genotype of the child [33].

While cross-sectional studies do not indicate whether parenting styles may affect substance use or vice versa, the analysis of longitudinal, cross-lagged effect sizes showed that the authoritative parenting style predicts a decrease or weaker increase of substance use, respectively, while associations of other parenting styles with change in substance use did not reach statistical significance. As only 12 to 17 cross-lagged effect sizes were available for the other parenting styles, we have to interpret the lack of statistical significance with caution. Thus, more studies are needed before final conclusions on statistical effects of these parenting styles on change in substance use can be drawn.

We found some support for the assumption that associations of parenting styles with substance use vary by country levels of collectivism/individualism, although the results supported only, in part, the suggestions of Garcia and Garcia [13]. As vertical individualism and vertical collectivism predicted stronger negative associations of authoritative parenting with substance use, the vertical dimension was predictive. Thus, there were stronger associations in cultures that accept hierarchical structures, such as parental control within the family. As no moderating effect of the assessed cultural dimensions were found for authoritarian parenting—which is also characterised by high control [6, 7]—the quality of parental control seems to play a role. While both authoritative and authoritarian parents use firm, direct, and consistent control, authoritarian parents also use coercive discipline, such as physical punishment and hostile criticism [34]. Thus, our data indicate that vertical collectivism and vertical individualism may only promote the effects of non-coercive parental discipline on substance use in the offspring.

While permissive parenting tended to be associated with higher levels of substance use in countries low of horizontal collectivism, there was, on average, no association in countries with high horizontal collectivism scores. Thus, there was no support for the assumption that permissive parenting may protect the offspring from substance use in horizontal collectivistic cultures (as suggested by [13]). Affective ties between parents and children were insufficient for preventing and/or reducing substance use if not accompanied with some parental control, although the low parental control of permissive parents seemed not to be a risk factor for substance use in countries with high horizontal collectivism scores, such as Brazil or Colombia and in studies that differentiated between permissive and neglectful parenting.

The observed stronger association of authoritarian parenting with use of nicotine as compared to alcohol may indicate that young people are more likely to cope with stress associated with authoritarian parenting by smoking cigarettes rather than drinking alcohol. For example, Corneau et al. [35] found that adolescents tended to have a stronger motive to use cigarettes as compared to alcohol for coping with stress.

Stronger associations of authoritative and permissive parenting with substance use in studies that used the same informant for assessing both variables may have been based on shared methods variance. Alternatively, as all studies with same informants had exclusively used offspring self-reports, stronger associations in these studies may also indicate that parenting is more likely to affect substance use if the parenting style is perceived by the offspring.

We did not find that associations of parenting styles with substance use decrease across adolescence and emerging adulthood. This indicates that parents may still provide some guidance in late adolescence and emerging adulthood or that representations of parenting styles that have been developed in the past and habits developed under the influence of the parents at an earlier age have a persistent influence on substance use across adolescence and emerging adulthood.

Neither child nor parental gender moderated the size of the association of parenting styles with substance use, thus indicating that mothers and fathers are similarly relevant for the substance use of their sons and daughters.

An important question refers to whether the present results might be biased. Most results on study quality

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indicate robustness of the present findings. There was no evidence for publication bias or bias due to selective reporting of findings. Although a substantial number of included studies had used non-validated parenting measures and only a minority of studies had confirmed a response rate of 80% or higher, we showed that these characteristics of study quality did not moderate the size of the reported effects. There was only one potential source of bias we were not able control for—the exclusive use of self-reports on substance use. It has been shown that adolescent self-reported substance use corresponds to results of urine analysis only in about 70% of the cases [36]. Relating parenting styles to physiological indicators of substance use would be relevant for a further test of the robustness of our findings.

6 | LIMITATIONS AND CONCLUSIONS

Some limitations of the present meta-analysis have to be acknowledged. First, most of the included studies analysed only concurrent associations of parenting styles with substance use and did, therefore, not allow for testing associations of parenting styles with change in substance use and vice versa. Thus, the analyses of cross-lagged effects provided only a preliminary picture. Second, fewer studies had assessed neglectful parenting compared to other parenting styles because some measures address only the three parenting styles introduced by Baumrind [6], such as the Parental Authority Questionnaire [27]. Third, although we included data from 45 countries across 6 continents, these countries represented only about 23% of the countries of the globe. In addition, data on the four aspects of collectivism/individualism were only available for a subset of the included countries [21]. Finally, the included studies assessed substance use only with self-reports. Future studies should add more objective measures, such as biological measurements for validating the associations of parenting styles with self-report measures of substance use.

Despite these limitations, the following conclusions can be drawn. First, parenting styles, and the authoritative and neglectful style in particular, are associated with substance use in adolescents and emerging adults. However, as the average associations are small and as cross-lagged effects are very small in a statistical sense, expectations about potential effects of parenting styles should not be too high. Second, there is some cultural variation in the size of associations of parenting styles with substance use, and young people may benefit most from authoritative parenting in cultures that appreciate some parental control and strictness. The different forms of collectivism and individualism proposed by Triandis and Gelfand [14] explain part of the contradictory results that appear in the literature. There is no empirical support for the assumption that a permissive parenting style may have positive effects on substance use in horizontal collectivistic societies, although the use of this style does not predict higher substance use in this cultural context.

Third, when being interested in effects of permissive parenting, researchers need to use measures that differentiate between permissive and neglectful parenting. Fourth, as the included studies represent only 23% of the countries of the globe, more studies are needed from the other countries to address which parenting styles act as a protective factor and which as a risk factor under which societal conditions.

Fourth, knowledge on associations of parenting styles with substance use can be used for planning parent-focused/family-focused prevention programs. A meta-analysis by Allen et al. [37] showed that parenting interventions are effective at preventing and decreasing adolescent substance use, although the focus was on parenting in general rather than on parenting styles in particular. As parenting styles are defined by the combination of different parental behaviours (firm control, warmth), future studies could test whether interventions addressing combinations of parental behaviours are more effective than those with a narrow focus on an individual parental behaviour.

Finally, additional research is recommended on longitudinal associations of parenting styles with change in substance use across the globe. Experimental studies—such as randomised controlled trials on effects of parenting training—should be conducted for testing causal hypotheses about effects of parenting styles. For understanding how parenting styles affect substance use, research is needed on mediators of the effects of parenting styles on substance use in the offspring.

AUTHOR CONTRIBUTIONS

Study conception and design: MP, MP and JL coded the studies, MP performed the statistical analyses and drafted the manuscript, MP and JL revised the manuscript. Both authors substantially contributed to the manuscript and approved the final version.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no competing interests.

DATA AVAILABILITY STATEMENT

Data on characteristics of the included studies is provided in the electronic supplement.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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